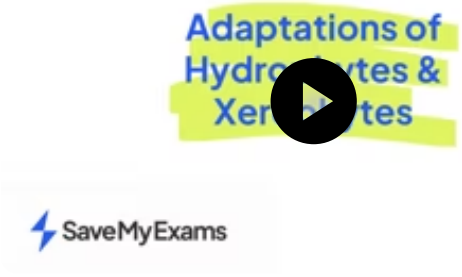
Adaptive Features Continued

Exam code: 0610 & 0970

Download PDF

Written by: Lára Marie McIvor
Reviewed by: Lucy Kirkham
Updated on 3 October 2024



Did this video help you? Yes No

Help me understand

Hydrophytes & Xerophytes: Extended

Extended Tier Only

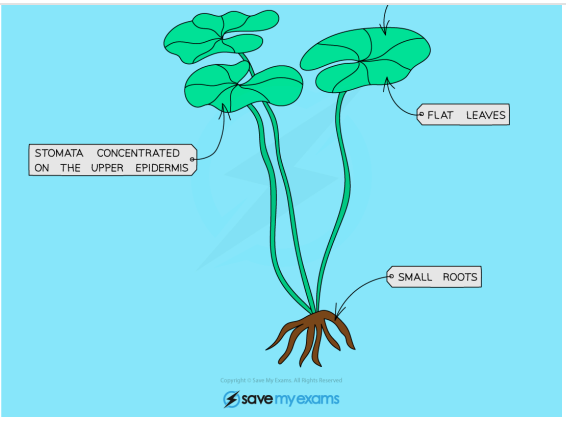
Hydrophytes

- Plants adapted to live in **extremely wet conditions**
- Common adaptations include:
  - **Large air spaces in their leaves** for flotation, to keep the leaves close to the surface of the water where there is more light for photosynthesis
  - **Small roots** as they can also extract nutrients from the surrounding water through their tissues
  - Stomata usually open all the time and **mainly found on the upper epidermis** of the leaf where they can exchange gases much more easily with the air

Stuck on something?

Ask questions and get help with exactly what you're revising.

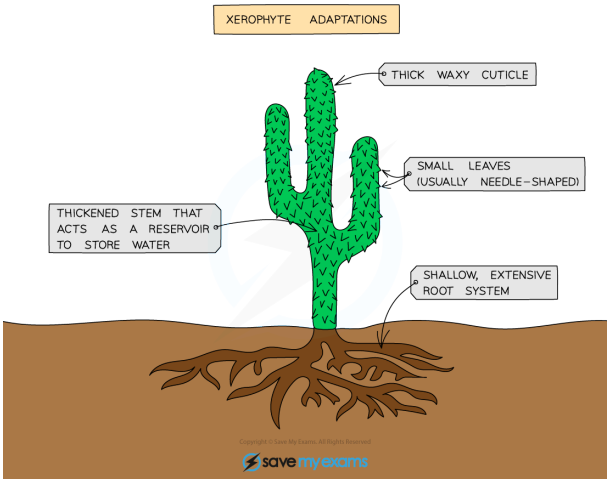
Got it



Hydrophytes are adapted to live in wet conditions such as ponds

Xerophytes

- Plant adapted to live in **extremely dry conditions**
- Common adaptations include:
  - **Thick waxy cuticle** - the cuticle cuts down water loss in two ways: it acts as a barrier to evaporation and also the shiny surface reflects heat and so lowers the temperature
  - **Sunken stomata**: stomata may be sunk in pits in the epidermis; moist air trapped here lengthens the diffusion pathway and reduces the evaporation rate
  - **Leaf rolled** with stomata inside and an inner surface **covered in hairs** - traps moist air and prevents air movement across stomata which reduces transpiration
  - **Small leaves**: many xerophytic plants have small, needle-shaped leaves which reduce the surface area and therefore the evaporating surface
  - **Extensive shallow roots** allow for the quick absorption of large quantities of water when it rains
  - **Thickened leaves or stems** which contain cells that store water



Xerophytes are adapted to live in extremely dry conditions such as deserts

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself Flashcards

Was this revision note helpful? Yes No

Previous Next

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.